Lab 2 - H-Bridge DC Motor Drive Experiment

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Abstract

Motors are widely used in robotics, industrial machinery, automation, and other applications. In this experiment, we constructed a circuit based on an *Arduino Uno R3* (ATmega328P) and an *L293D* motor driver chip to control a DC motor. We demonstrated **speed adjustment** and **direction reversal** of the motor. Additionally, by utilizing light-dependent resistors (LDRs) to modulate the motor's behavior, we proposed a **light-sensitive** motor control system as a practical application.

Introduction

This experiment aims to demonstrate the fundamental principles of DC motor control using an H-bridge circuit. The specific objectives include:

- Constructing a motor driver circuit with an *L293D* chip and an *Arduino Uno* microcontroller.
- Adjusting the motor speed by varying the input voltage (via PWM signals).
- Reversing the motor direction by switching the polarity of the H-bridge outputs.

To achieve these goals, we programmed the Arduino to generate control signals for the *L293D*, which acts as an H-bridge to drive the motor. The results show that the motor speed can be linearly controlled by the analog voltage applied to the potentiometer pin (pin 0), while the direction is toggled by a digital signal on the switch pin (pin 7). This setup also extends to light-sensitive control by replacing the potentiometer with LDRs.

Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

Method

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

Results

Add all appropriate figures and analysis. No formal word count for this section, however please keep text as concise as possible.

Conclusions

Please give the conclusions from the characterisations including a summary of the results collected.